# 1 Introduction

In that laboratory number 02, purpose of the labrotary is applying **Superposition Theorem**, **Norton** and **Thevenin equivalents**. We will find **Norton** and **Thevenin** equivalents and **Superposition Theorem** on circuits. Then we will use CircuitLab web Labrotary Simulation to check if our calculations are correct.

## 1.1 Preliminary Work

1. Find Thevenin and Norton equivalent circuits between node A and B of the circuit shown in Figure 1. Using the equivalent circuits calculate the voltage between A and B.

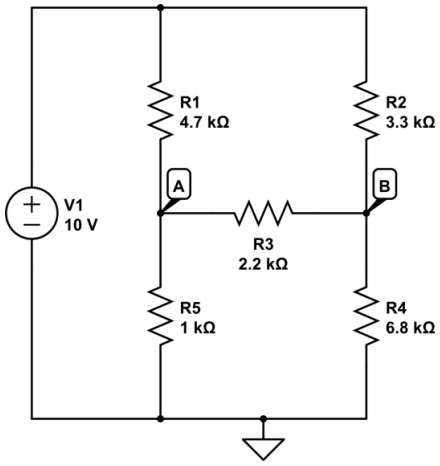



Figure 1: Task 1 Thevenin and Norton equivalents

2. Use superposition to find the current flowing through R4 in Figure 2.

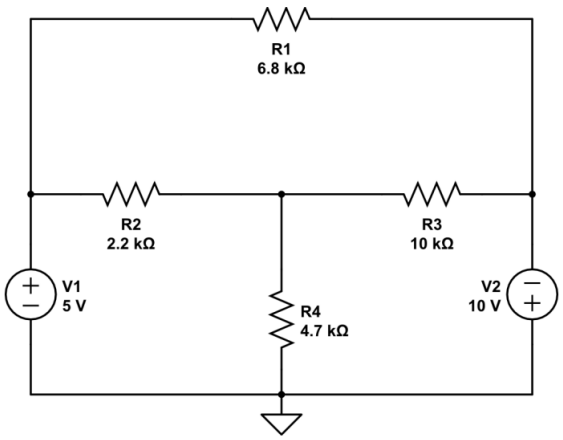



Figure 2: Task 2 Superposition Theorem

## 1.2 Preliminary work solutions

### 1.2.1 Solution for Task 1, (1.1.1)

1. On the circuit which stated on Figure 1, we will find Norton and Thevenin equivalents to calculate voltage between node A and B. First we are calculating the current passing through the circuit if we connect node A and B ignoring R3.

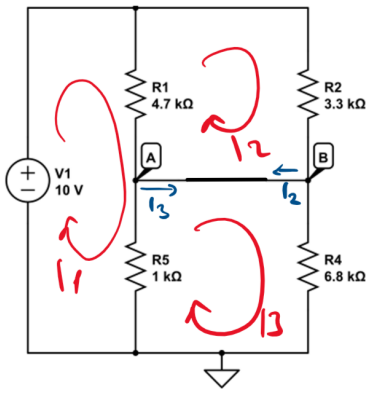



Figure 3: Preliminary work 1 Norton Equivalent

2. We draw current ways on each loop and stated their currents as I1 I2 and I3. We selected all of them on Clock-Wise (CW) direction to calculate easily. Here we apply Mesh Analysis to calculate current passes through from all of the loops in Figure 3

3. From the Figure 3, we can calculate 3 different equations to find Norton Equivalent. Here we use In*R = V formula to find out equations with Kirchoff Voltage Law, which states on closed loop there will be 0 total voltage. From first loop we have:
I1 * 4.7k + I1 * 1k - I2 * 4.7k = 10V
From second loop we have:
I2 * 4.7k + I2 * 3.3k - I1* 4.7k = 0
From third loop we have:
I3 * 1k + I3 * 6.8k - I1 * 1k = 0
and if we transform that equations into matrix we will have the Figure 4.

$$\begin{pmatrix} 5.7\,k & -4.7k & -1k \\ -4.7k & 8k & 0k \\ -1\ k & 0k & 7.8k \end{pmatrix} \begin{pmatrix} I1 \\ I2 \\ I3 \end{pmatrix} = \begin{pmatrix} 10V \\ 0 \\ 0 \end{pmatrix}$$

Figure 4: Matrix form of 1.2.1.3

4. I used python to solve that matrix *(Code is stated on Figure 5)* and I had:
I1 = 0.00355803 we round it to 0.00356
I2 = 0.00209034 we round it to 0.00209
I3 = 0.00045616 we round it to 0.00046
We have I2 (0.00209) on node B and I3 (0.00046) on node A, if we subtract I3 from I2 we will find the current passing through the line between node A and B.
That equation will be 0.00209 - 0.00046 = 0.00147. Here 0.00147 is current between node A and B. That is our Norton Equivalent.

```python
import numpy as np

# Define the coefficient matrix and the constant vector
k = 1000
Reistances = np.array([[5.7*k, -4.7*k, -1*k],
                       [-4.7*k,   8*k,    0],
                       [-1*k,     0,  7.8*k]])

Voltages   = np.array([10, 0, 0])

# Solve the system of linear equations
x = np.linalg.solve(Reistances, Voltages)

print(x)  # Prints the solution vector
```

```
IPython Console
Console 1/A
Python 3.8.10 (tags/v3.8.10:3d8993a, May  3 2021, 11:48:03)
[MSC v.1928 64 bit (AMD64)]
Type "copyright", "credits" or "license" for more information.

IPython 8.8.0 -- An enhanced Interactive Python.

In [1]: runfile('C:/Users/seyit/untitled0.py', wdir='C:/Users/
seyit')
[0.00355803 0.00209034 0.00045616]
```

Figure 5: Python code solves the matrix on Figure 4, Currents are listed in order.

5. Next we will find Thevenin with opening the circuit as stated on Figure 5.

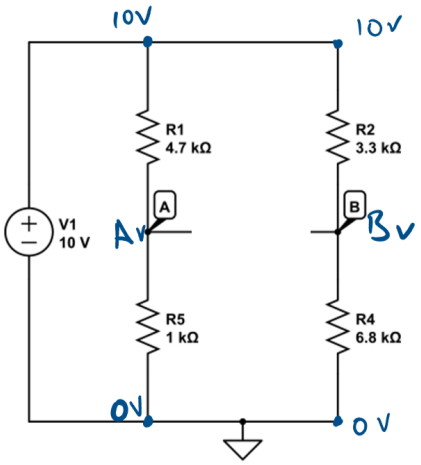



Figure 6: R3 is removed from circuit to calculate Thevenin Open Circuit Voltage.

6. We have 10 Volt DC voltage supply for all of the circuit here, and we need to calculate voltage difference between node A and B, so we name the node points as Av and Bv. To calculate voltages, we are separating 10 volt to 2 separate ways to feed circuit. First way is concluding R1 and R5 and the second way is concluding R2 and R4. We have 10 Volts on the beginning of the ways and 0 Volts on the ends. So basically we are separating 10 volts to two different way rationally with resistance of ways.
   so we have
   Av/1k = 10/5.7k with that calculation we have Av is nearly equal to 1.754 volts.
   Bv/6.8k = 10/10.1k with that calculation we have Bv is nearly equal to 6.733 volts.
   And that is our Thevenin Equivalent.
   So here we have **Av (1.754)** and **Bv (6.733)** we are calculating voltage between node A and B. We are subtracting Av from Bv *(subtraction way is only showing the direction of current)* so we have:
   **6.733 - 1.754 = *4.979 Volt between Node A and B.***

### 1.2.2 Solution for Task 2, (1.1.2)

1. find current passes through R4 we will use **Superposition Theorem.** We will find it with two steps, first we will ignore V2 then V1.

2. First we are removing V2 to use Superposition Theorem.

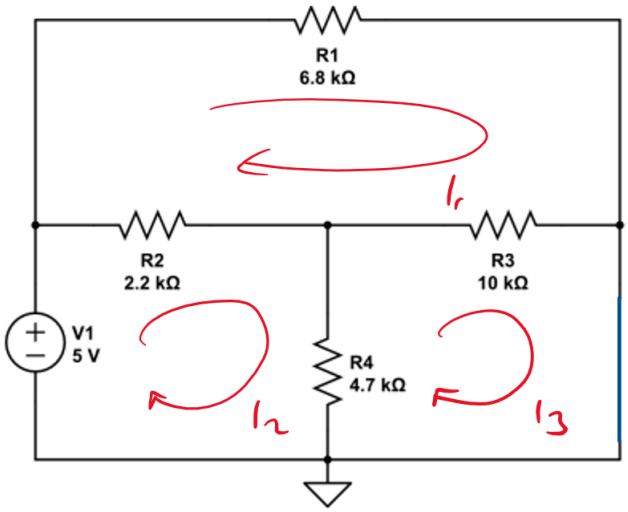



Figure 7: Step 1 of Superposition Theorem. V2 removed from Figure 2 for Task 2

3. As we used before Mesh Analysis on Task 1 (1.1.1), we are using Mesh Analysis again with same steps.
   Repeating the 1.2.1.3,
   Here we have:
   I1 * 19k - I2 * 2.2k - I3 * 10k = 0
   I2 * 6.9k - I1 * 2.2k - I3 * 4.7k = 5V
   I3 * 14.7k - I2 * 4.7k - I1 * 10k = 0
   If we transform that three equations into matrix we will have:

$$\begin{pmatrix} 19\text{ k} & -2.2\text{k} & -10\text{k} \\ -2.2\text{ k} & 6.9\text{k} & -4.7\text{k} \\ -10\text{ k} & -4.7\text{k} & 14.7\text{k} \end{pmatrix} \begin{pmatrix} I1 \\ I2 \\ I3 \end{pmatrix} = \begin{pmatrix} 0 \\ 5V \\ 0 \end{pmatrix}$$

Figure 8: Matrix form of 1.2.2.3

   And if we solve that matrix with same code we used on 1.2.4, and shown in Figure 5 we have:
   I1 = 0.00073529 we are rounding up to 0.00074
   I2 = 0.00166169 we are rounding up to 0.00166
   I3 = 0.00103149 we are rounding up to 0.00103

4. We are looking for current passing through the R4 so we have to subtract I3 (0.00103) from I2 (0.00166) to have current.
   So we have **0.00166 - 0.00103 = *0.00063 Ampere*** *Current passing through R4* for step 1 of Superposition.

5. On the next step now we are ignoring V1.

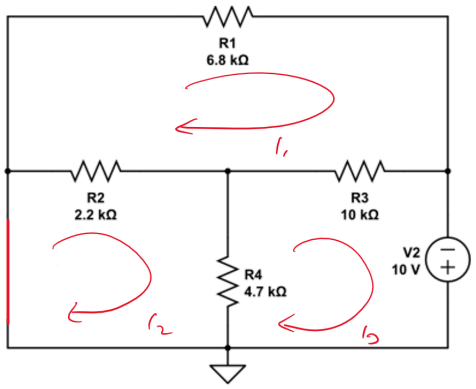



Figure 9: Step 2 of Superposition Theorem. V1 removed from Figure 2 for Task 2

6. If we calculate three equations for three currents we will see the nearly exact same equations from 1.2.2.3 but there is a voltage difference it is not from second equation it is from third equation and it is not 5 Volts it is 10 Volts now
   Here we have:
   I1 * 19k - I2 * 2.2k - I3 * 10k = 0
   I2 * 6.9k - I1 * 2.2k - I3 * 4.7k = 0
   I3 * 14.7k - I2 * 4.7k - I1 * 10k = 10V
   If we repeat the step 1.2.2.3, we will have matrix looks like on Figure 10.

$$\begin{pmatrix} 19\text{ k} & -2.2\text{k} & -10\text{k} \\ -2.2\text{ k} & 6.9\text{k} & -4.7\text{k} \\ -10\text{ k} & -4.7\text{k} & 14.7\text{k} \end{pmatrix} \begin{pmatrix} I1 \\ I2 \\ I3 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \\ 10V \end{pmatrix}$$

Figure 10: Matrix form of 1.2.2.6

7. And from the calculation with same code shown in Figure 5. We will have:
   I1 = 0.00147059 we are rounding up to 0.00147
   I2 = 0.00206298 we are rounding up to 0.00206
   I3 = 0.00234026 we are rounding up to 0.00234

8. We are looking for current passing through the R4 so we have to subtract **I2 (0.00206)** from **I3 (0.00234)** to have current.
   So we have **0.00234 - 0.00206 =** ***0.00028 Ampere*** *Current passing through R4* for step 2 of Superposition.

9. Now we have two different currents passing through the R4, we need to add them up to find total current passing through the R4
   we have 0.00063 Amp from step 1 and 0.00028 Amp from step 2. If we add them up we will obtain **total current passing through the R4 which is** ***0.00063 + 0.00028 = 0.00091 Amp***

## 1.3 Lab Tasks

In the laboratory we will check if our calculations are working in simulation. We will build exact same circuits on the Circuit Lab Simulations and check if the Voltages (1.2.1.6) and Currents (1.2.2.9) are correct in our calculations.

### 1.3.1 Task 1

1. We will construct Figure 1 on the Circuit Lab.

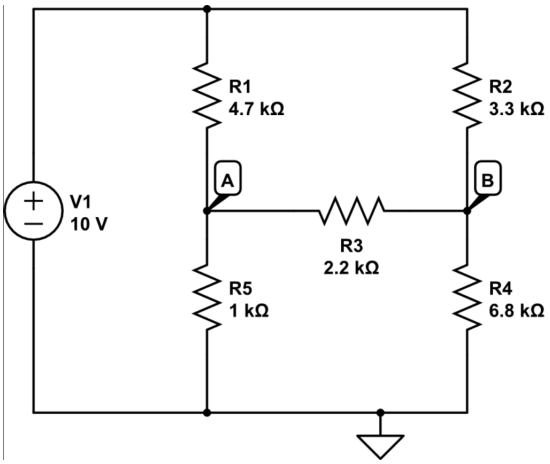



Figure 11: Task 1, circuit on Circuit-Lab and named nodes A and B.

2. Now we are going to measure voltage between nodes A and B, which contains R3 (2.2k).

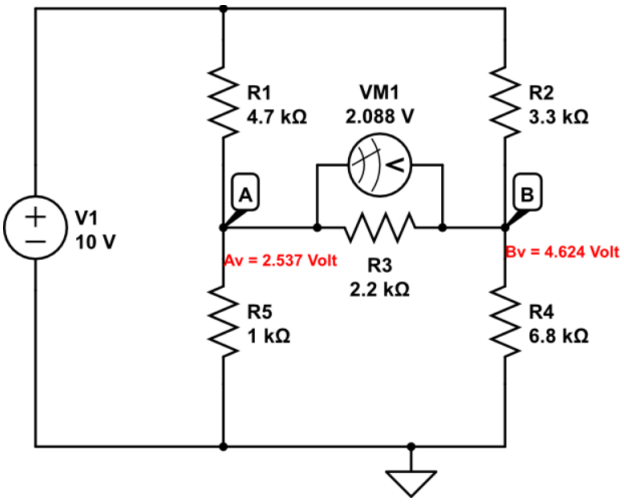



Figure 12: Voltage measured between nodes A and B.

3. We have measured voltages on node A and named as Av, and on node B and named as Bv. Here we have 2.537 Volts on Av and 4.624 Volts on Bv. Difference between two nodes is equal to ***4.624 - 2.537 = 2.087 Volts.***

Table 1: Expected and Calculated values for 1.3.1.3

| | V | I | R |
|---|---|---|---|
| Expected | 04.979 | 0.00147 | 2.2k |
| Calculated | 2.088 | 0.949 | 2.2k |

4. Now we are removing R3 (2.2k) and calculating Thevenin Open Circuit Voltage, *this step is same with 1.2.1.6*,

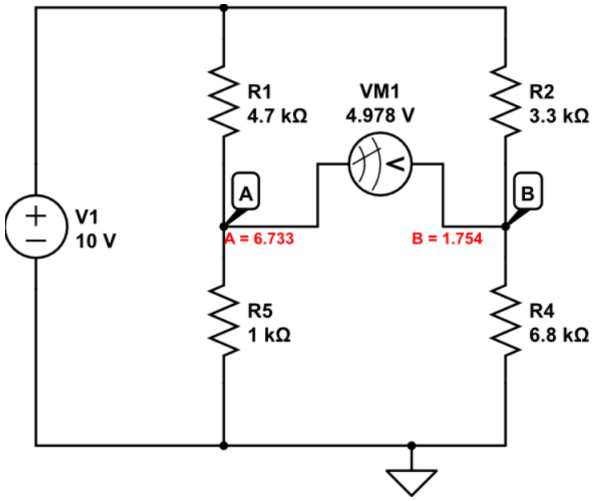



Figure 13: Open circuit for 1.3.1.4, R3 is removed from circuit to calculate *Thevenin Open Circuit Voltage*

So here we have **Av (1.754)** and **Bv (6.733)** we are calculating voltage between node A and B. We are subtracting Av from Bv *(subtraction way is only showing the direction of current)* so we have:
**6.733 - 1.754 = *4.979 Volt between Node A and B.***
And we are adding little amount of resistance between node A and B to analyze voltage drop between two nodes.

5. Now we are, short circuiting nodes A and B together and measuring the short circuit current.

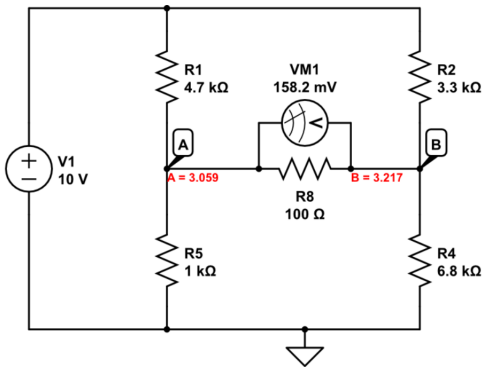



Figure 14: R8 100 ohm Resistor is added between node A and B

With that calculation Node **A has 3.059** Volts and **B has 3.217** Volts, if we subtract A from B we will have ***3.217 - 3.059 = 0.158 Volts*** **between node A and B. So that from Ohm's Law *( V = I*R )* 0.158 / 100 = I, I = 0.00158 Amp.**

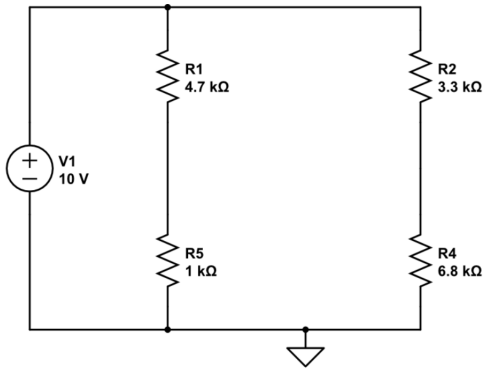



Figure 15: R3 removed from circuit to find out Thevenin Open Circuit Resistance.

6. Measurements from 1.3.1.5 and 1.3.1.4 we have 4.979 Volts and 0.000158 Ampere.

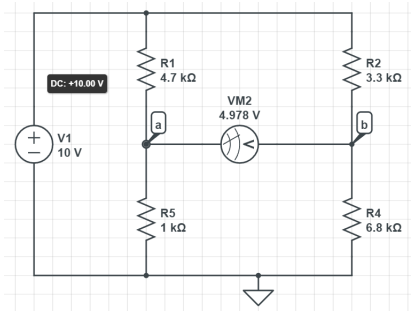



Figure 16: Open circuit for 1.3.1.4, R3 is removed from circuit to calculate *Thevenin Open Circuit*

We had 4.978 Volt between node a and b. That is **Vopen**.

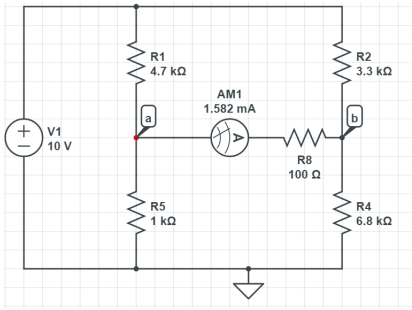



Figure 17: Short circuit for 1.3.1.4, *Norton Short Circuit*

We had 1.582 mAmp between node a and b. That is **Ishort**.

7. with Vopen and Ishort we can find Resistance, Vopen = Ishort * Resistance
Resistance = Vopen / Ishort =¿ Resistance = 4.978 / 1.582 so that ***Resistance is 3.1466 ohm***.

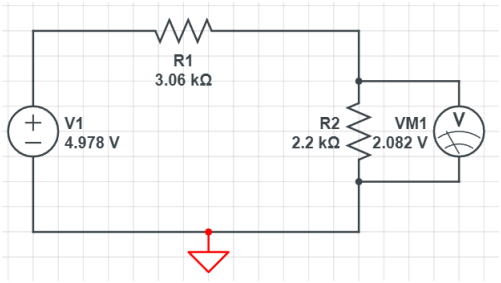



Figure 18: Simplified version of Figure 11

8. V open circuit. It is equal to 4.987V. Then create a short circuit by using a very little amount of resistance (10 ohms is used) and measure the current flowing through that resistor. It is equal to 1.629mA. Then calculate R thevenin by ohms law. It is 4.987/(1.629*100**-3) = 0.304k ohms. Finally link r thevenin and r load to the V open and measure the V on rload. It is 2.082V which is very close to the expected 2.088V.

## 1.4 Task 2

1. In this task we are looking to find current passes through R4. on Figure 19.

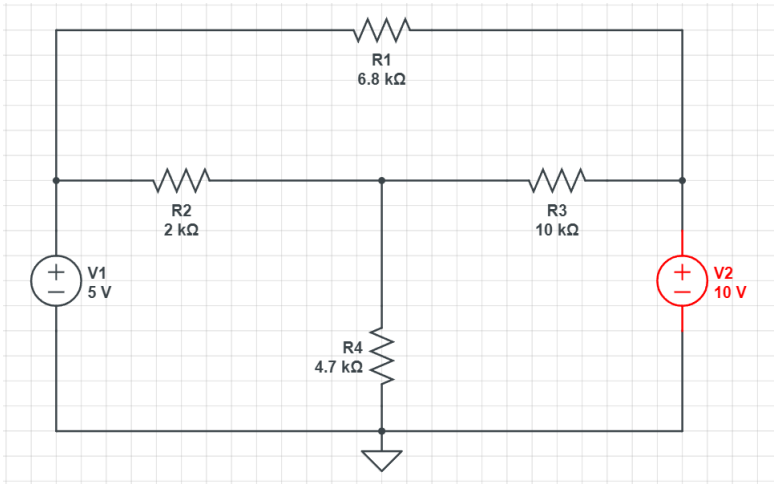



Figure 19: Superposition circuit for 1.4.1

2. first we are removing V1 (5V) and calculate the current passing through the R4.

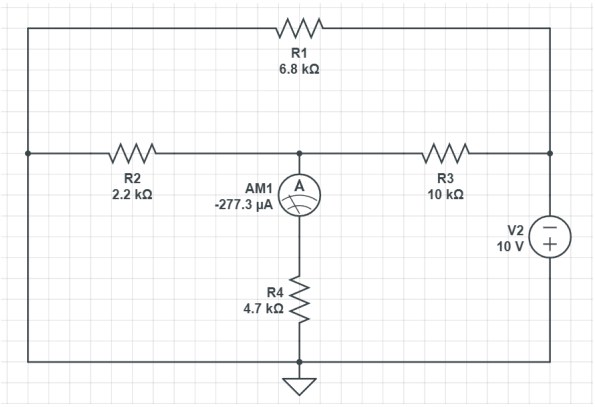



Figure 20: V1 is removed from Figure 19.

3. with that step we had 277.3 uAmp passes through the R4 in opposite direction as the base statement.

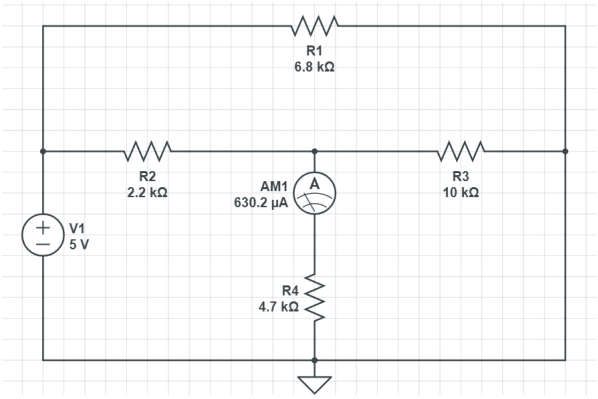



Figure 21: V2 is removed from Figure 19.

4. with that step we had 630.2 uAmp passes through the R4.

5. with steps we have used in preliminary work and steps 3,4,5 and 7 we from 1.3.1. we can calculate same values.
   At the end we need to add two different values comes from 1.4.3 (-277.3 uAmp) and 1.4.4 (630.2 uAmp) and we can get ***352.9 uAmp.*** *That is current flows through the R4*

# 2 Conclusion

Finding the potential difference between nodes a and b using Thevenin and Norton circuits, with R4 acting as the load resistor, was the first job in this circuit analysis challenge. This was accomplished by taking R4 out of the circuit and substituting it with an open circuit, which allowed us to calculate V open circuit as the potential difference between nodes a and b. Next, Isc and Rthevenin were calculated using Ohm's law and a very modest resistance linked between nodes a and b. We were able to compute Ieq using these data to design a simple circuit. The Superposition Theorem was used to the second task's objective, which was to determine the current flowing through R4.
This is my URLs: 1. 2. 3. 4. 5.